COA QUESTION BANK FOR END SEMESTER EXAMS UNIT WISE

UNIT-I COA Important Questions

1	Explain the basic structure of a computer with a neat block diagram and describe the functions of each unit.
2	Explain system bus structure including address bus, data bus and control bus.
3	Using 2's complement method, perform the following subtractions: **(Data may be changed)** i) 1010100 − 1000011 ii) 100101 – 110011
4	Explain error detection using parity bits. Distinguish between even parity and odd parity with suitable examples.
5	Draw the odd parity generator circuit and explain its operation.
6	Draw the Flow Chart for signed-magnitude Addition and Subtraction and explain its operation.
7	Explain the architecture and working of an Array Multiplier with neat diagram.
8	Describe Booth's multiplication algorithm with a neat flowchart.
9	Perform **(+6) × (−5)** using Booth Algorithm**. (Data may be changed)**
10	Explain restoring division algorithm and describe its hardware architecture.
11	Determine 1011 ÷ 101 using restoring division. **(Data may be changed)**

UNIT-II COA Important Questions

1	Explain the RTL concept in digital system design.
2	Describe the common bus system using multiplexers with help of neat diagram.
3	Describe the common bus system using three state buffers with help of neat diagram.
4	Draw the binary adder–subtractor circuit and explain its operation.
5	Describe logic microoperations. Explain any four logic microoperations with examples.
6	Explain shifting microoperations including logical, circular, and arithmetic shifting operations.
7	What is instruction code and explain the stored program concept.
8	Describe the computer registers of a basic computer and state the function of each register.
9	Explain the basic computer instruction formats and classify different types of instructions.
10	Draw the flowchart of instruction cycle and explain.
11	Describe the interrupt cycle and explain how an interrupt is handled in the basic computer.
12	Describe the basic architecture of the computer and its internal components.

UNIT-III COA Important Questions

1	Explain the General Register Organization in a basic computer with the help of block diagram.
2	Illustrate PUSH and POP operations in stack organization.
3	Explain Three, Two, One and Zero addressing formats.
4	Compare RISC and CISC architectures.
5	Classify instructions based on functionality and discuss data transfer and arithmetic instructions.

6	Classify instructions based on functionality and discuss logical and shift instructions.
7	Illustrate any five addressing modes.
8	Explain Microprogrammed control organization.
9	Discuss the concept of address sequencing with the help of block diagram.
10	Explain the fields of Microinstruction format.

UNIT-IV COA Important Questions

1	Discuss the memory hierarchy in computer system.
2	Explain the hardware organization of associative memory with the help of block diagram.
3	Illustrate the mapping techniques in cache memory.
4	Describe the features of virtual memory and explain the address translation in virtual memory
5	Explain how the CPU is connected to main memory (RAM and ROM) with a neat diagram.
6	Explain the asynchronous data transfer modes.
7	Illustrate the “Daisy Chaining method” and “Parallel Priority Interrupt method” of interrupt handling.
8	Explain the Direct Memory Access (DMA) mode of data transfer with a neat diagram.
9	Explain the Programmed I/O mode of data transfer with a neat diagram.
10	Describe the DMA controller and its operation with a neat diagram.

UNIT-V COA Important Questions

1	Discuss the features of multiprocessor systems.
2	Explain interconnection structures in multiprocessor systems.
3	Illustrate the time-shared common bus and multiport memory organization.
4	Explain the serial (daisy chain) arbitration technique with a neat diagram.
5	Explain the parallel arbitration technique using priority encoder and decoder.
6	Explain pipelining concept with examples.
7	Describe the instruction pipeline with its stages and explain with an example.
8	Explain the RISC pipeline and discuss its operation with an example.
9	describe the attached array processor using suitable diagram.
10	Describe the organization of a SIMD array processor with a neat diagram.